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Ivar Cashin Eriksson

Dear Members of the Selection Committee,

I am writing to apply for the PhD scholarship in Uncertainty Quantification for Large Language Models at DTU Compute. I have a Master's degree in Statistical Learning and Data Analytics from KTH, a background in Engineering Physics, and seven years of experience applying machine learning in research and production settings. My background is strongest in probabilistic modelling, model evaluation, and production ML systems, and I would like to develop that into research on how language models should express uncertainty.

My strongest research-adjacent experience comes from RaySearch Laboratories, where I worked on automated radiotherapy planning using deep learning, probabilistic modelling, and optimisation. There I built latent representations from segmented 3D CT scans with autoencoders and predicted dose-volume histograms and spatial dose distributions using sparse Gaussian processes. The domain was not language modelling, but it taught me to work with uncertainty in a high-stakes setting and to treat evaluation as part of the modelling problem rather than an afterthought.

I have also worked in applied settings where model behaviour needed to be understood clearly. At Valcon, I built a production real-time neural network on natural-language inputs and high-dimensional labels, and I led explainable AI and Git trainings. At Signum Life Science, I currently work on a forecasting platform for pharmaceutical price and demand forecasting, and I use LLMs to embed free text for similarity search and matching workflows. That work is adjacent to the advertised topic, but not the same as studying uncertainty in LLMs directly, which is why the PhD is attractive to me.

The topic also fits my academic background. At KTH, I specialised in Statistical Learning and Data Analytics, with a broader engineering foundation in mathematics and physics. Relevant coursework included Statistical Machine Learning, Probability Theory, and Optimization. I also spent a year at ETH studying Applied Mathematics. That training is the main reason I am comfortable moving back towards a more research-focused role.

DTU Compute looks like a strong environment for this work because the topic is both mathematical and applied. I would value the chance to study how uncertainty can be measured, calibrated, and used in large language models in a way that is useful to researchers and end users alike. I do not claim direct prior work in LLM uncertainty methods, but I do bring experience with probabilistic modelling, model evaluation, and production ML systems, together with the motivation to develop that into PhD research.

I would welcome the opportunity to discuss my background and how it could contribute to the project. Regards,

Ivar Cashin Eriksson.